

Rover-Based Autonomous Science by Probabilistic Identification and Evaluation

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Abstract Autonomous science augments the capabilities of planetary rovers by shifting the identification and selection of science targets from remote operators to the rover itself. This shift frees the rover from wasteful idle time and allows for more selective data collection. This paper presents an approach to autonomous science that is comprised of three components: a Bayesian network that uses image data to identify features; an evaluation algorithm that selects the best features; and, a path-planning algorithm that guides the rover to the most scientifically valuable features. Within this framework, the effectiveness of pair-

ing a larger prime rover with a smaller scout rover to improve autonomous science is investigated. Laboratory-based experiments were used to validate the effectiveness of the Bayesian network for feature identification and the scoring algorithm that has been developed for feature evaluation. Simulations were used to compare the traditional use of a solo prime rover to that of also employing a scout. The results presented here indicate that the inclusion of a scout rover can allow the prime rover to avoid pitfalls or routes with low scientific value.

Keywords Autonomous science · Planetary robotics · Bayesian networks

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1 Introduction

The guidance of a robotic vehicle can be inefficient if it is dependent on human-selected goals. This is particularly true if the operator has limited access to the robot (e.g., during the exploration of hostile environments such as in disaster areas, war zones, remote regions on Earth or other planets). In the scenario of planetary exploration, automatic target selection is typically motivated by the perceived scientific value of observed

features. Automatically extracting this scientific information (e.g., identifying minerals in a rock field) and using it to influence the behaviour of the planetary rover is known as *autonomous science*.

The need for this type of system is apparent when one considers, for example, the operating conditions of rovers on Mars. Opportunities for communication with operators on Earth are very infrequent, usually limited to one telemetry/command cycle per sol (Martian day). In addition, bandwidth limitations restrict the amount of data that rovers can send back to Earth. This bottleneck is particularly restrictive because current (e.g., Curiosity) and future (e.g. ExoMars) rovers are larger, have greater range, and carry more scientific sensors than their predecessors. As a result, it is necessary that rovers become more selective in their data collection. They may be required to autonomously select only the best and/or most representative data for transmission back to Earth.

To fully address the issues described above, an autonomous science system must be generally capable of: (i) detecting, identifying, and localizing scientific targets; (ii) evaluating identified features based on their scientific value and the goals of the mission; and, (iii) mission planning to accommodate path or destination changes based on evaluated features. There have been relatively few efforts to date that have effectively performed all three of these steps, and virtually no system has extensive heritage, due in part to the scarcity and cautious nature of rover missions.

In many research efforts, partial implementations of autonomous science systems have been developed. Bayesian classification has been used to autonomously detect meteorites in Antarctica [21] and rocks in analogue Martian terrain [20] (without an evaluation or path-planning component). More ambitious systems have included components from all aspects of autonomous science. Systems including detection, evaluation and re-planning components have been developed and evaluated in laboratory environments [5, 22]. However, these systems did not include feature identification (using attribute measurements to estimate the type of rock) or a machine learning component—the potential value of including such a component was noted in [22].

1.1 About this Paper

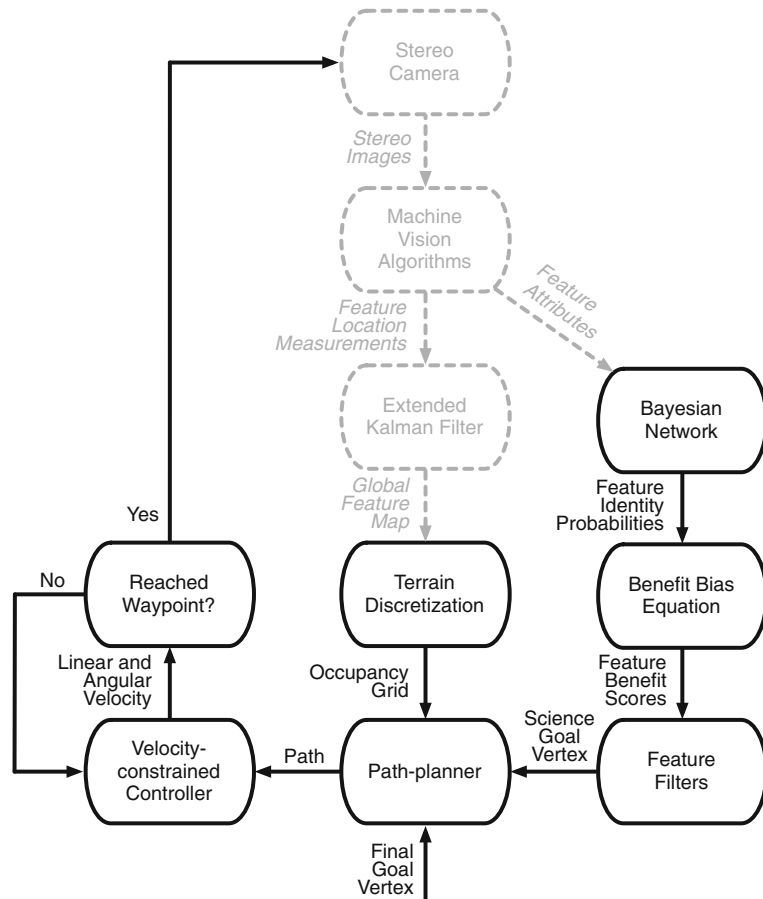
This paper presents a Bayesian approach to autonomous science that includes aspects of the three capabilities outlined in Section 1. Feature detection, localization, and attribute extraction are provided by image processing techniques performed on images captured by a stereo-camera. A Bayesian network (BN) is trained to identify objects based on their attributes. Next, identified objects are evaluated considering the confidence of their identification, the quality of the measurement, and the current goals of the rover. Finally, a planner considers both the science reward and detour required to visit objects before planning paths. The purpose of this paper is to study the combination of these elements at a high-level and to make observations about how autonomous science and path-planning could be intertwined into a mutually beneficial control system. A specific possible application of the system (the scenario involving a prime and scout rover) forms the platform for these observations.

The entire autonomous science system described in this paper was tested and evaluated in simulation under two scenarios: (i) prime rover solo; (ii) prime/scout rover pairing. An example of a prime rover is ExoMars (a large rover currently under development by the European Space Agency) and of a scout rover is Kapvik (a small, rugged rover that was developed in part by Carleton University). The effectiveness of using a small, rugged, inexpensive scout rover to aide the larger and more conservative prime rover in performing autonomous science was evaluated. The results presented here indicate that the inclusion of a scout rover could allow the prime rover to avoid significant pitfalls or routes with low scientific value. Finally, laboratory-scale testing using a small mobile robot was performed to validate the use of the BN as a practical object identifier.

2 Algorithm Architecture

The overall architecture of the developed autonomous science system is illustrated in Fig. 1. This paper focuses on the identification and evaluation of features, the feature selection criteria, and

Fig. 1 The general architecture of the developed autonomous science system. This paper focuses on those elements illustrated with *solid lines*



the planning and execution of science-influenced paths. What follows is a general description of the algorithms (the nodes in Fig. 1) that make up the system.

1. A stereo camera captures an image of the environment and, for each detected feature, machine vision algorithms generate an estimated position and a value of each attribute being measured.
2.
 - a. For each detected feature, a BN calculates the probability that the feature is a particular object, based on the attribute measurements (e.g., feature 1 has object probabilities of {0.14, 0.63, 0.09} for objects A, B, and C, respectively).
 - b. Using the feature position estimates, an extended Kalman filter (EKF) adds new features to the feature map or updates the position of previously observed features.
3.
 - a. The benefit bias equation (BBE) assigns a benefit score to each of the identified features based on its object probabilities, apparent size and the current mission goals of the rover.
 - b. To prepare for graph-based path-planning, the terrain is discretized into an occupancy grid based on the feature map.
4. The scored features are filtered based on their benefit scores and the detour that is required to visit them. Features that pass through the filters are ranked based on their “score-to-detour” ratio. The highest ranked feature is nominally selected as the current science goal.
5. Using the occupancy grid, a path is planned to the current science goal using Theta* [15] path-planning. If there is no current science goal, the goal is the final goal originally specified by the operator at the beginning of the traverse.

6. The path is supplied to a velocity-constrained controller [3] that generates velocity commands for the rover and the rover then drives on the planned path. Once a waypoint is reached, return to Step 1.

Detailed descriptions of Steps 1 and 2b are not provided in what follows, since these subjects lie outside the scope of this paper. However, the output they provide is briefly described in Section 2.2. The remaining steps are all described in greater detail in Section 3.

2.1 Localization Assumptions

An autonomous science system requires the integration of many different algorithms from a variety of fields. Transforming the range and bearing measurements of features (provided by stereo vision) into a global feature map requires knowledge of the pose of the rover that captured the image. Although an extended Kalman filter (EKF) was implemented to update the estimated position and uncertainty of features, rover localization was assumed. Such algorithms that perform simultaneous localization and mapping (SLAM) are widely available, including EKF SLAM, unscented Kalman filter (UKF) SLAM and FastSLAM—a review of these techniques is described in [7]. Also, multiple observations of the same feature would need to be correlated using data association, which has been developed for use with SLAM [2]. Overall, a SLAM algorithm that includes a data association component would be a good approximation of the information gained from the assumptions made above.

2.2 Machine Vision Work

The machine vision and feature position estimation components of an autonomous science system are not presented here, since these lie outside the principal focus of this paper. However, these functions have been developed and were used in generating the results presented in this paper. The data provided by the machine vision algorithms was made available from the output of collaborative work within the authors' research group. Briefly, image segmentation used k -means cluster-

ing [13] and the measured textural attributes are a subset of the Haralick Textural Parameters [10]. The attributes measured in the laboratory experiments presented in this paper were size, intensity, contrast, energy, entropy and inverse difference moment (IDM). The estimated range and bearing of features was calculated using stereo triangulation. When it was required, an EKF was implemented to update the estimated position of features in the global reference frame, based on the errors and uncertainties of the triangulation process and the assumed knowledge of the pose of the rover.

3 Algorithm Description

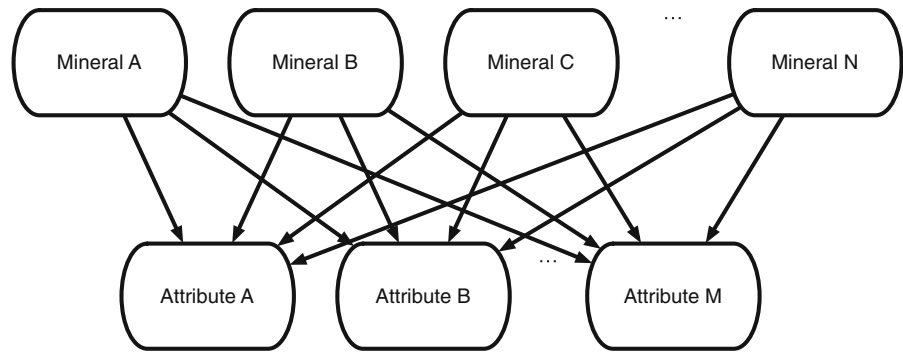
This section provides detailed descriptions of the steps of the autonomous science system outlined in Section 2. As already mentioned, Steps 1 and 2b, which lie outside the focus of this paper, are briefly described in Section 2.2.

3.1 Bayesian Network

A generalized model of the Bayesian network (BN) [6] used in this paper is shown in Fig. 2. Note the causality of the directed acyclic graph (DAG); the N minerals *cause* the states of the M attributes. For simplification, the BN was limited to discrete rather than continuous variables. This assumption reflects the type of observations a geologist would make in the field, as geologic features are usually identified qualitatively (e.g., dark, pitted, layered, etc.) rather than quantitatively (e.g., 36 % colour intensity, 19 % pit coverage, average bedding thickness of 4.4 mm). Every mineral has M child nodes and every attribute has N parent nodes. Minerals have two states (true or false) and attributes $A, B, C, \dots, M$ have $m_A, m_B, m_C, \dots, m_M$ states.

The BN is trained using a set of F features, each of which has an associated M measurements. As the measurements that provide the evidence to update the probabilities of each node are usually continuous (e.g., a measurement of the size of a rock), fuzzy logic [23] is used for discretization. The membership functions for the states of each attribute are determined by analyzing the distrib-

Fig. 2 The generalized model of the BN used in this paper



ution of measurements for that attribute from the training set. This process is demonstrated in Fig. 3. The resulting training data set is F sets of measurements of attributes $A, B, C, \dots, M$ attribute that are each fuzzified into $m_A, m_B, m_C, \dots, m_M$ states. For example, a measurement of attribute A might be $A = 134.23$. Using the trapezoidal parameters of the state membership functions (such as those shown in Fig. 3), this continuous measurement can be described as degrees of truth pertaining to each possible state. If A has states $\{low, medium, moderate, large\}$, $A = 134.23$ might be mapped to $A = \{0, 0, 0.34, 0.58\}$. This partial membership to different states is used as *soft evidence* for the BN.

Bayes' Theorem (BT) is used to initialize the conditional probability tables (CPTs) for each

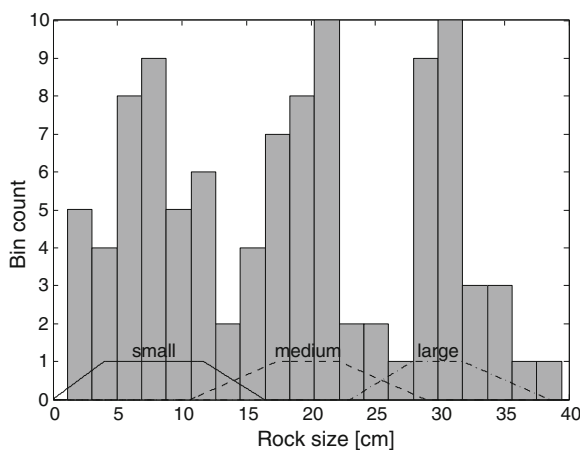


Fig. 3 An example of selecting the fuzzy membership functions of the attribute states based on the training results. Here, three trapezoidal functions map continuous measurements to membership of {small, medium, large}

node in the BN, which specify the conditional probabilities of a child node (the attributes) based on the state of its parent node(s) (the minerals). For example, for two minerals (α and β) and two attributes ($A = \{small, medium, large\}$ and $B = \{low, medium, high\}$), the CPT for attribute A might resemble Table 1. The probabilities in the final column are calculated using the entries in the training data set. For example, the probability $\Pr(A = small|\alpha)$ (i.e., the probability in the first row in the table) is calculated as

$$\Pr(A = small|\alpha) = \frac{\Pr(\alpha|A = small)\Pr(A = small)}{\Pr(\alpha)}. \quad (1)$$

Now that the conditional probabilities of all child nodes are known, BT can be used to updated the mineral probabilities when a new measurement is introduced after it is fuzzified into soft evidence.

3.2 Benefit Bias Equation

A benefit score (S_b) is assigned to each feature that was identified by the BN using the *benefit bias equation* (BBE), which considers the value of discovering different mineralogies based on

Table 1 An example of a CPT

α	β	A	$\Pr(A \alpha, \beta)$
True	False	Small	0.63
True	False	Medium	0.19
True	False	Large	0.05
False	True	Small	0.02
False	True	Medium	0.82
False	True	Large	0.11

the operator-selected overall mission of the rover. The BBE takes the form

$$S_b = K \cdot P \cdot \Sigma \cdot [a\Pr(A) + b\Pr(B) + \dots + n\Pr(N)], \quad (2)$$

where K is the scale constant, P is the apparent size (also known as angular diameter) of the feature, Σ is the normalized standard deviation of the probabilities, and $a, b, \dots, n$ are the importance weights for minerals $A, B, \dots, N$. The scale constant K simply scales scores to be in an appropriate scale. It does not affect the relative scores of different features. The apparent size P of a feature is

$$P = 2 \tan^{-1} \left(\frac{r}{d} \right), \quad (3)$$

where r and d are the estimated radius and distance to the feature, respectively. The apparent size is measured in radians. It reduces the score of features that have had only a small number of pixels analyzed compared to features with larger numbers of pixels, as less information was available for the machine vision algorithms. Another purpose of P is to determine whether S_b is to be updated. Each time a feature is observed, S_b is updated if P is larger than any previous observation. In general, this means that S_b represents the benefit score calculated when the most visual information is available about that feature.

The normalized standard deviation Σ adjusts S_b based on the standard deviation of the set of probabilities $\{\Pr(A), \Pr(B), \dots, \Pr(N)\}$. When there is only unreliable information about a feature (such as a distant observation), the different mineral probabilities are often very similar, as no distinguishing attributes can be measured. The purpose of Σ is to raise the score of features whose probabilities are more distinct and have been narrowed down to one or two minerals; i.e.,

$$\Sigma = \frac{\text{StDev}\{\Pr(A), \Pr(B), \dots, \Pr(N)\}}{\text{StDev}\{i_1, i_2, \dots, i_N\}},$$

$$i_k = \begin{cases} 1 & \text{if } k < \frac{N}{2}, \\ 0 & \text{if } k \geq \frac{N}{2}. \end{cases} \quad (4)$$

The denominator of (4) simply normalizes the standard deviation of the probabilities.

The importance weights $a, b, \dots, n$ for minerals $A, B, \dots, N$ adjust the benefit score based on the current mission of the rover. For example, if mineral A was highly desirable, importance weight a would be given a high value. Conversely, if mineral B was undesirable, b would be given a smaller value relative to a . Note that b could be assigned a negative value if finding mineral B detracts the rover from visiting an area, while assigning b a value of zero would ignore this type of mineral all together.

3.3 Terrain Discretization

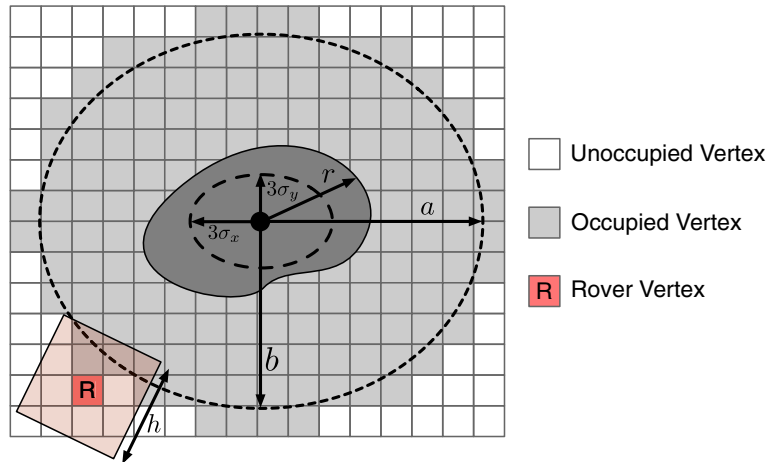
Discretization of the terrain is required so that a graph-based path-planner can be used. In this implementation, the map created via discretization is a binary occupancy grid. Path-planning on this type of map attempts to minimize the distance travelled between start and goal vertices. Vertices are occupied if they reside within the area of influence (A_j) of a feature j . The rover maintains a list of features it has observed with their estimated positions and covariances. The area of influence considers the estimated radius of the feature (r_j), the standard deviation of its position ($\sigma_{x,j}$ and $\sigma_{y,j}$), and the maximum dimension of the rover's footprint (h). The area of influence is an ellipse with semi-major (a_j) and semi-minor (b_j) axes

$$a_j = r_j + 3\sigma_{p,j} + \frac{h}{2}, \quad (5a)$$

$$b_j = r_j + 3\sigma_{q,j} + \frac{h}{2}, \quad (5b)$$

where $p = x$ and $q = y$ if $\sigma_{x,j} > \sigma_{y,j}$, or $p = y$ and $q = x$ if $\sigma_{x,j} < \sigma_{y,j}$. The estimated radius r_j is conservatively selected as the half of its estimated largest dimension. The axes are expanded by half of h so that the rover can be represented by a single vertex during path-planning (Section 3.5). An illustration showing how the occupancy status of vertices is determined is shown in Fig. 4. All vertices that fall within or are touched by the boundary of the area of influence A_j of a feature are considered occupied.

Fig. 4 The area of influence of a feature is a function of the feature's radius (r), the uncertainty in its position (σ_x, σ_y) and largest dimension of the rover (h)



Points of scientific interest in the feature map must also be represented on the discretized map. The “science vertices” of a feature are all unoccupied vertices surrounding that feature that are within the science radius (r_{sci}), defined as

$$r_{\text{sci}} = r + r_{\text{sen}}, \quad (6)$$

where r is the estimated radius of the feature and r_{sen} is the radius of the science instrument being used for up-close observation.

3.4 Feature Filtering

Evaluated features are filtered to eliminate those with insufficient benefit scores or require too great a detour to perform up-close observation. Features passing through the filters are ranked based on their score-to-detour ratio.

The benefit score filter uses both static and dynamic thresholds to eliminate features based on their benefit scores. The static threshold simply eliminates features below a set benefit score, while the dynamic threshold depends on the standard deviation of the benefit scores of all the features currently in the map. The active threshold ($S_{b,\text{min}}$) depends on the criteria

$$S_{b,\text{min}} = \begin{cases} S_{b,\text{stat}} & \text{if } m_D \mu < S_{b,\text{stat}}, \\ m_D \mu & \text{if } m_D \mu \geq S_{b,\text{stat}}, \end{cases} \quad (7)$$

where $S_{b,\text{stat}}$ is the static threshold, μ is the mean benefit score of the feature set, and

$m_D = (\mu + \sigma)/\mu$, making the dynamic threshold one standard deviation (σ) above the mean.

The detour filter eliminates features that would require an excessive detour for visitation for up-close observation. In this paper, the rover is assumed to be on a mission with a set final goal and has limited energy to reach that goal. Calculating the path length of a detour requires calculating the path length (L) to every science vertex (Section 3.3) of a feature and from every science vertex to the goal. The selected science vertex $s_{q,\text{sci}}$ for feature q is the science vertex that satisfies

$$\min [L_{s_{\text{start}} \rightarrow s_{q,i}} + L_{s_{q,i} \rightarrow s_{\text{goal}}}], \text{ for } i = 1, 2, \dots, \eta \quad (8)$$

where $L_{s_a \rightarrow s_b}$ is the path length from vertex s_a to s_b and η is the set of science vertices for feature q . Features passing through the detour filter must satisfy the inequality

$$L_{s_{\text{start}} \rightarrow s_{q,\text{sci}}} + L_{s_{q,\text{sci}} \rightarrow s_{\text{goal}}} \leq D L_{s_{\text{start}} \rightarrow s_{\text{goal}}}, \text{ for } D \geq 1 \quad (9)$$

where D is the detour coefficient. The detour coefficient could be selected using several different methods. For simplicity, a static D is used in this paper; however, using a dynamic D that changes depending on energy availability may be advantageous.

Features that have passed through both the benefit score filter and the detour filter are now ranked by their score-to-detour ratio S_b/L_D ; i.e.,

$$\frac{S_b}{L_D} = \frac{S_b}{L_{s_{\text{start}} \rightarrow s_{q,\text{sci}}} + L_{s_{q,\text{sci}} \rightarrow s_{\text{goal}}}}, \quad (10)$$

which specifies the amount of scientific value the rover will get per unit distance travelled. The feature with the greatest S_b/L_D is selected as the best target for up-close observation.

3.5 Path-Planning

All autonomous planetary rovers to date have used graph-based path-planning algorithms. This includes Sojourner [9], Spirit and Opportunity [4], and Curiosity [1]. One advantage of using graph-based path-planning is that it is guaranteed to find a solution if such a solution exists. In this paper, a binary occupancy grid is used for path-planning. Once the rover is localized to a vertex in this map and a goal vertex is specified, the path planner plans the optimal path (i.e., the shortest length) from the rover vertex to the goal vertex. A widely used algorithm to plan graph-based paths is A* [11], from which many modern algorithms have been derived. A major drawback of using A* path-planning is restricting movement to be along the edges connecting the occupied vertex and its neighbouring vertices. In an eight-connected grid (each vertex has eight neighbours in a grid pattern), this means the heading of a path is limited to increments of $\frac{\pi}{4}$. To remove this restriction, an any-angle derivation of A* called Theta* [15] is used in this paper. Theta* allows path-planning between any two vertices with line-of-sight (i.e., if it is possible to connect them without crossing any occupied vertices).

3.6 Velocity-Constrained Control

To test the autonomous science algorithm, a suitable control policy for the rover is required. The kinematic controller used in this paper is the path-following, velocity-constrained controller developed by Bak et al. [3]. This controller has two key properties that make it ideal for the rover

model in this paper executing Theta*-planned paths. First, it allows for the specification of several types of velocity constraints, namely: minimum and maximum vehicle forward velocities, minimum and maximum vehicle angular velocities, minimum and maximum wheel angular velocities, and a desired vehicle forward velocity. The controller attempts to minimize the lateral and heading error from the path while maintaining the desired forward velocity but not violating any of the velocity constraints. Second, because the path is known, the controller can anticipate upcoming turns and adjusts the velocities such that desirable turn properties (overshoot, settling time, etc.) are achieved, once again without violating velocity constraints. The combination of these two properties allows for smooth, constrained execution of the planned paths.

4 Experimental Methods

This section describes the two environments that were used to test the autonomous science algorithm outlined in Section 3. A laboratory test environment was used to validate using a BN for object identification and to investigate the effectiveness of the BBE for feature evaluation. To test the autonomous science algorithm in its entirety, a simulated environment was developed and tested in MATLAB and used to compare two different rover scenarios.

4.1 Laboratory

A Point Grey Bumblebee XB3 CCD [17] stereo camera was mounted on a Pioneer 3-AT mobile robot [14] on a static mast at a height of 1.17 m from the ground, tilted downward at an angle of 30°. Features were measured with respect to the left camera. The mobile robot simply acted as a “mobile tripod”. No data was used from its sensors and it was driven with a joystick. Localizing the robot (and hence, the camera) in the global reference frame was performed via manual measurements each time an image was captured. The stereo camera mounted on the mobile robot is shown in Fig. 5.



Fig. 5 Point Grey Bumblebee XB3 CCD Camera mounted on the Pioneer 3-AT mobile robot

A BN was used to identify the 13 different objects listed in Appendix B. Evidence from six different attributes (size, intensity, contrast, energy, entropy, IDM) was used. The BN used a directed acyclic graph of the generalized form illustrated in Fig. 2. The BN was first trained with 30 test images that contained one of each of the 13 objects (although not all objects were detected in every image). One of these images is shown in Fig. 6. Note that images were converted to grayscale before being processed, so colour could not be used to distinguish between objects. Although using colour would have greatly facilitated classification, the stark differences in colour among the objects does not reflect the spectrum one might expect on a planetary mission. The trained BN was used to identify features in four test trials. In each of these trials, the rover was driven approximately 10 m through a field of 30 features, capturing 9–15 images along the way. The 30 features were comprised of six small brown pots, six small green pots, three white plastic balls,

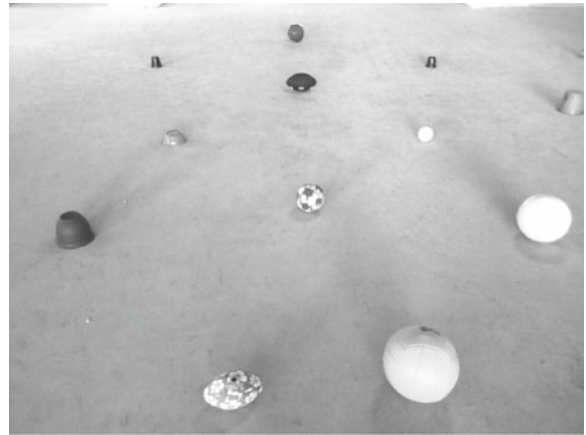


Fig. 6 One of the 30 rectified, grayscale images used to train the BN

three yellow plastic balls, four bowls, and one of each of the remaining eight objects specified in Appendix B. Objects were randomly placed in 30 different static feature locations before each trial. For feature evaluation, a scale constant of $K = 10$ and the importance weights listed in Table 2 were used in the benefit bias equation (Section 3.6). The results of the laboratory experiments are presented in Section 5.1.

It is worth noting that using artificial objects could both positively and negatively affect the identification process. As colour was not used for identification, the variety of shapes, sizes and textures of the objects made them reasonable proxies

Table 2 The importance weights used in the laboratory experiments

Object	Importance weight
Pink volleyball	−1
White volleyball	+1
Small soccer ball	−1
Large soccer ball	+3
Bowl	−3
Small football	−1
Large football	−1
White plastic ball	+3
Yellow plastic ball	−1
Small green pot	+1
Small brown pot	+1
Medium pot	−3
Large pot	−1

for rocks. Making identification more difficult is the fact that some objects became visually similar in grayscale (e.g. the two volleyballs, the two small pots). Also, many artificial objects lack unique texture signatures that could help distinguish them from similar but differently composed objects. Conversely, identifying two of the same object becomes simpler as their features are more likely to match. Finally, the vector of measured attributes was somewhat limited and only included data from a single camera. On a planetary rover, it is much more likely that additional sensors will be available that would aid the BN in distinguishing and identifying objects. It is, however, noted that the final system should operate in natural terrain. This process is underway and is noted in Section 6.2. Overall, it was felt that the laboratory scenario properly tested the preliminary effectiveness of the BN in object identification because of the challenges described above.

4.2 Simulator

A simulated rover, modelled after a differential drive vehicle, was driven through a randomly created rock field with rock size and density approximating Martian terrain (modelled after a proposed ExoMars landing site [18]). Rocks over 10 cm in diameter were modelled as insurmountable obstacles in the occupancy grid. In each trial, a simulated data set of features was created that was used to train a BN. Machine vision data was emulated to provide the rover with estimated feature positions and to provide the BN with attribute measurements. The fundamental purpose of the simulator was to test the full integration of the parts comprising the autonomous science system. It also demonstrates how the data obtained in the laboratory experiments could be used in two different yet meaningful scenarios: the prime rover solo and a prime/scout rover pairing.

Prime Rover Solo: The prime rover solo scenario consists of a single rover driving through the terrain to a final goal that is 10 m from its starting position. This is the general scenario described by the steps outlined in Section 2 and the diagram in

Fig. 1. The goal in this scenario is for the rover to reach its final goal having visited a feature for up-close observation (i.e., occupied one of its science vertices) along the way, while maximizing the score-to-detour ratio.

Prime/Scout Rover Pairing: The prime/scout rover pairing is a modification of the prime rover solo scenario. Here, the scout rover drives to the 10 m goal while mapping its local environment. This map includes the location of all observed features as well as their estimated mineralogy probabilities. After arriving at the goal, the scout sends this information to the prime rover, which then plans a path to the feature with the highest score-to-detour ratio. The prime rover then drives to this feature for up-close observation before heading to the final goal.

The prime/scout rover pairing scenario described above was selected on the basis that it demonstrates how autonomous science may influence path-planning (and vice-versa) by providing scientific information to the prime rover as it plans a medium-length traverse. This data-sharing cooperation provides *a priori* obstacle and science information to the better-equipped and more valuable prime rover at the expense of greater energy consumption and risk by the scout rover. Data transfer between entities on the Martian surface has previously been used effectively. The Sojourner rover communicated with the Pathfinder lander via radio modems at a rate of 2.4 kbps with a maximum range of 500 m [12]. It is assumed that modern rovers would greatly exceed this data rate due to the relatively limited capabilities of Sojourner and Pathfinder. Scenarios that divide the labour differently between the rovers and the accompanying effect on performance of the system could be investigated in future work. The results of the simulator experiments are presented in Section 5.2 and the parameters that were used in the simulations are provided in Appendix A.

5 Results and Analysis

This section outlines the results of the two types of testing that were performed to evaluate the

autonomous science algorithm; namely, the laboratory experiments and the simulator. The results of the laboratory experiments are described in Section 5.1, focusing on the performance of the feature identification and feature evaluation algorithms presented in Sections 3.1 and 3.6, respectively. In Section 5.2, the results from the simulator are presented and analyzed, with a focus on comparing the two scenarios described in Section 4.2.

5.1 Laboratory

The laboratory experiments explored the utility of training a BN (Section 5.1.1) for feature identification (Section 5.1.2) and the effectiveness of the evaluation algorithm (Section 5.1.3).

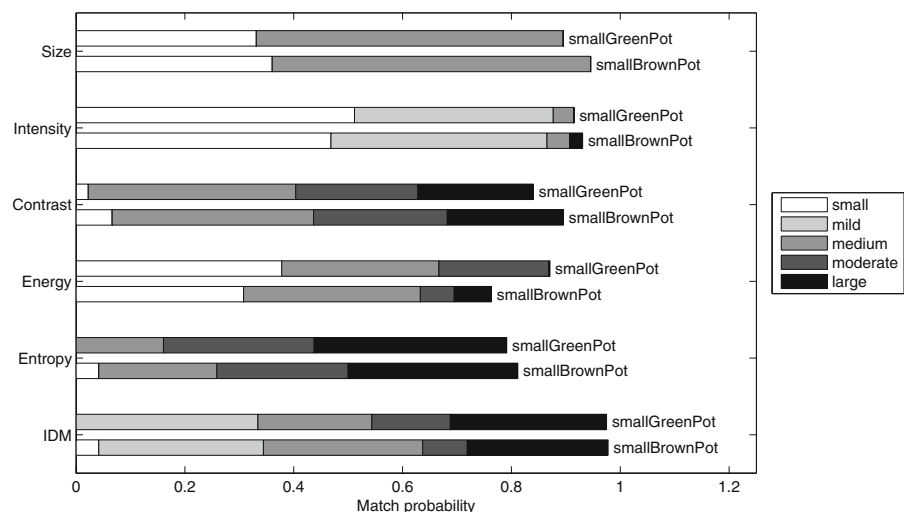
5.1.1 Training

The BN used a data set of 30 images to be trained using the method described in Section 3.1. However, not every object was properly segmented and detected by the machine vision algorithms in every image. Overall, 171 objects were identified over the 30 images. Histograms of the continuous measurements for each attribute were used to specify the trapezoidal parameters of the fuzzy membership functions, using the process illus-

trated in Fig. 3. Size, contrast, energy and entropy were each divided into four states: small, medium, moderate and large. Intensity and IDM had a fifth state (mild) between small and medium. Similarities in the membership values among some of the objects provide insight into possible difficulties in differentiating objects during the test trials. If state memberships are too similar for most or all the attributes between two objects, the BN may split the probability between the similar objects. Two objects that shared particularly similar state memberships were the small green pot and small brown pot (Fig. 7).

The omission of colour as an attribute (for the reasons described in Section 4.1) can cause objects similar in size and texture have similar state memberships. The differences between the two small pots were minute for all six attributes. This leads to the prediction that upon observing either of these objects during the test trials, it will be impossible for the BN to differentiate between them and a split in probability is most likely. If differentiating these objects were critical, it is likely that at least one additional attribute is required (with colour being an obvious choice). Other objects sharing resemblances had similar results. This observation emphasizes the importance of selecting attributes that measure a variety of physical characteristics of an object. For

Fig. 7 A comparison of the *small green pot* and *small brown pot* training results. Note that only intensity and IDM have a “mild” state



example, an infrared camera could be used to measure the thermal capacities of different rock types that appear physically similar.

5.1.2 Feature Identification

Over the course of the four test trials, 194 features were detected with machine vision and identified using the trained BN. It should be noted that false positives detected by the machine vision algorithm (e.g., an empty patch on the floor) are not included in this total. The purpose for this omission is that differentiation between features and benign terrain is primarily the responsibility of machine vision, which is outside the scope of this paper. Theoretically, the BN could be also be used for this purpose by including non-features (e.g., sky, shadows, sand, etc.) in the training process.

The two highest-match percentages for each object are shown in Appendix C. Overall, the BN identified the correct object as the most probable match in 9 of 13 cases. However, as described below, the failure of the BN to correctly identify some of the objects is a direct result of the performance and consistency of the machine vision algorithms. In two of the four mismatches, the second most probable match was the correct object. Among the best identified objects were the white volleyball, large football and white plastic ball (Fig. 11b, g, and h, respectively). Two factors that were found to aide in the correct identification of objects are consistency and uniqueness. For consistency, the measured attributes of an object must be similar over several measurements and in different environments. This means test measurements will be more likely to resemble measurements taken during the tuning process. Consistency is primarily a function of the quality of the sensors and algorithms used to determine attribute measurements. The measurements of the white volleyball and white plastic ball were quite consistent due to the quality of their segmentation. Inconsistencies can occur when objects are improperly segmented causing only part of the object to be measured, or a patch of the background is segmented as part of the object. Because these two objects were easily segmented due to their contrast compared to the background, the full object itself was usually detected and yielded consis-

tent attribute measurements. For uniqueness, the state membership functions that describe an object must not closely resemble other objects. This reduces ambiguity between objects and allows for high probability matches. Also, dependencies between related attributes need to be indicated by the DAG of the BN. Methods revealing these dependencies can be inferred by test data [19].

Among the poorest performing objects were the small football and the yellow plastic ball (Fig. 11f and i, respectively). The top two most probable matches were incorrect for both of these objects. In both cases, there were issues with consistency between the training data and the test data. Also, compared to well segmented objects, the training set was small for these objects as the frequency of their detection was small. As was predicted when analyzing the training results (Section 5.1.1), there were difficulties in distinguishing objects that had similar attribute state memberships (particularly the two small pots). As is shown in Fig. 11j and k, the small green pot and small brown pot were more-or-less not differentiable. In both cases, the average match probability of the small green pot was slightly higher than the small brown pot. As state membership similarities are likely to occur when dealing with real rocks, this problem must be addressed. One suggestion is the inclusion of an additional sensor, such as a spectrometer or infrared camera, such that more diverse measurements of similarly appearing rocks can be obtained. A comparable result occurred when identifying the pink volleyball and white volleyball (Fig. 11a and b, respectively). In this case, identifying the pink volleyball resulted in a near even match probability between the two. However, the white volleyball was correctly identified with an extremely high match percentage, with the pink volleyball making up the remainder. This was likely due to the exceptional consistency of the attribute measurements for the white volleyball.

Overall, features that were consistently detected and measured by the machine vision algorithms were generally successfully identified by the BN. The misidentified features (4 of 13) suffered from inconsistent segmentation prior to being identified by the BN. To improve identification performance, refining the segmentation algorithm used to detect features and adding the

Table 3 The top two and bottom two scored features from the four laboratory test trials

Trial	Score	True object	P	Σ	Top two matches	
					First	Second
1	3.78	LSB	0.061	0.865	LSB (97.5 %)	LFB (2.5 %)
	1.58	WPB	0.136	0.962	WPB (99.9 %)	WVB (0.1 %)
	⋮	⋮	⋮	⋮	⋮	⋮
	−1.49	LPT	0.157	0.978	PVB (98.5 %)	WVB (1.5 %)
	−5.36	SBP	0.516	0.503	MPT (53.2 %)	LFB (46.8 %)
2	3.29	LSB	0.122	0.946	LSB (96.3 %)	LFB (3.7 %)
	1.06	SGP	0.181	0.585	SGP (67.5 %)	SBP (32.5 %)
	⋮	⋮	⋮	⋮	⋮	⋮
	−4.26	YPB	0.318	0.530	BWL (77.3 %)	YPB (15.4 %)
	−5.93	MPT	0.618	0.571	LFB (65.9 %)	MPT (34.1 %)
3	16.63	MPT	0.627	0.937	LSB (95.8 %)	LFB (4.2 %)
	4.04	WVB	0.483	0.928	WVB (95.1 %)	PVB (4.9 %)
	⋮	⋮	⋮	⋮	⋮	⋮
	−2.53	SBP	0.243	0.503	MPT (53.3 %)	LFB (46.7 %)
	−2.59	BWL	0.222	0.591	MPT (77.2 %)	LSB (14.1 %)
4	4.74	WPB	0.222	0.771	WPB (95.5 %)	BWL (2.4 %)
	4.01	LFB	0.142	0.967	LSB (97.8 %)	LFB (2.2 %)
	⋮	⋮	⋮	⋮	⋮	⋮
	−5.16	SGP	0.513	0.500	MPT (50.7 %)	LFB (49.3 %)
	−8.80	MPT	0.752	0.576	MPT (75.9 %)	LSB (12.2 %)

capability to measure additional attributes of visually similar targets would be of great benefit. Additionally, to add robustness against false measurements, the BN could also be trained to identify benign featureless areas (e.g., the sky or unpopulated patches of sand).

5.1.3 Feature Evaluation

The top two and bottom two scoring features in each of the four trials are shown in Table 3. An example of a feature map generated by the rover is illustrated in Appendix D. As was indicated by the importance weights (Table 2), the large soccer ball and white plastic ball were given the greatest value. Moderate value was given to the white volleyball, small green pot and small brown pot. The pink volleyball, small soccer ball, small football, large football, yellow plastic ball and large pot were given moderately negative value, while the bowl and medium pot were considered the least valuable.

In three of the four trials, objects with the greatest value were assigned the highest benefit scores. In the exception (third trial), the object (a low-valued medium pot) was observed to have a relatively high apparent size and normalized standard deviation, but its misidentification as a highly-valued object (large soccer ball) was the primary cause of its bloated score. This exception highlights the importance of the two factors that aide in identification: consistency and uniqueness (discussed in Section 5.1.2). There were difficulties in producing consistent attribute measurements for the medium pot, which can lead to a set of measurements that more closely resembles another object. This error can be alleviated if the uniqueness of the measurements is improved by introducing new attributes or improving the diversity of the existing ones. However, the high scoring features were generally well identified and had benefit scores that reflected their importance.

In general, the lowest scoring features were often misidentified and had lower match

probabilities that were distributed among a greater number of objects. This highlights the importance of the normalized standard deviation, which lowers the benefit scores of features with ambiguous identities. It is also worth noting that as only the extremes in the scoring are shown in Table 3, outliers sometimes took precedence. For example, the small brown and green pots were generally well identified (although there was some ambiguity between the two). However, three of the four trials featured these pots as one of the lowest scoring features despite their moderate value. These mis-identifications were in fact outliers, as the small pots were generally identified as such in over 80 % of observations (Fig. 11j and k in Appendix C).

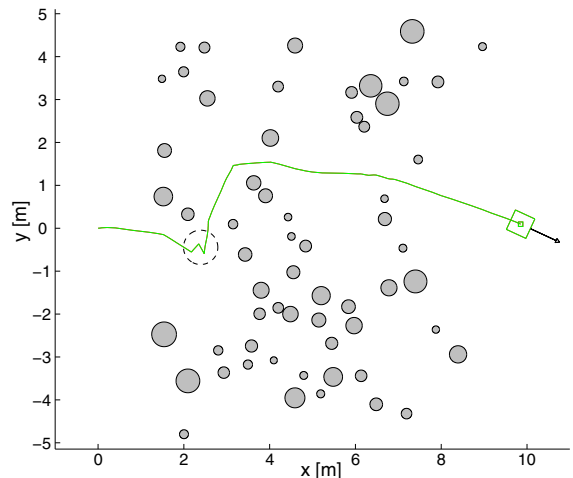
5.2 Simulator

The experiments using the MATLAB simulator enabled general observations of the behaviour of the two scenarios (Section 5.2.1) that caused a quantifiable difference in the results between them (Section 5.2.2).

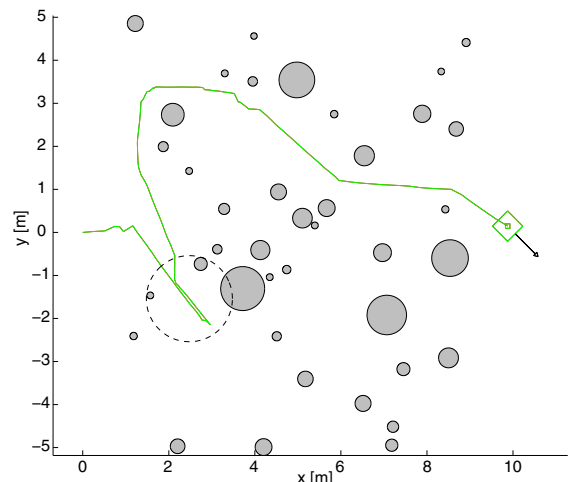
5.2.1 General Observations

Observations of the behaviour of the two scenarios provided a comparison of their strengths and weaknesses. These characteristics relate directly to the quantitative results presented in Section 5.2.2. Figures 8 and 9 illustrate examples of the behaviour observed for the prime rover solo and prime/scout rover pairing scenarios, respectively.

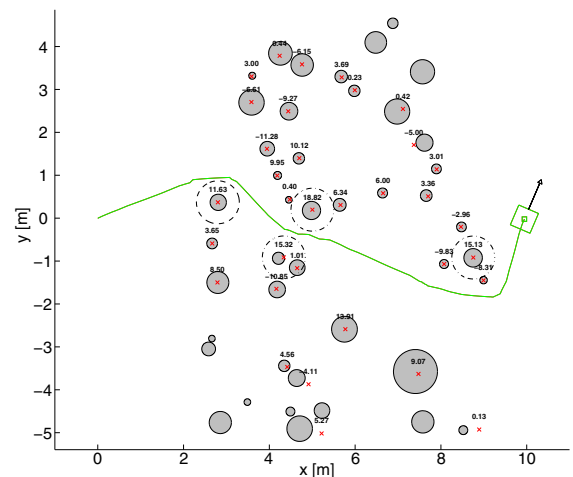
The disadvantage of having no *a priori* information about the features was apparent in some of the prime rover solo trials. One common issue was the appearance of *kinks* in the paths executed by the prime rover (Fig. 8a). Kinks were found to occur due to one of two reasons: indecisiveness in the feature selection or closures in the occupancy grid. Both reasons were caused by updates of the feature map. Indecisiveness in the feature selection occurred when the top ranking feature



(a) An example of a kink in the path of the prime rover solo is shown in the dashed circle.

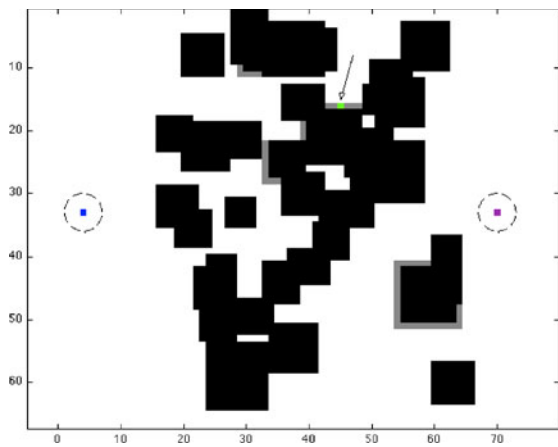


(b) An example of backtracking by the prime rover solo is shown in the dashed circle.

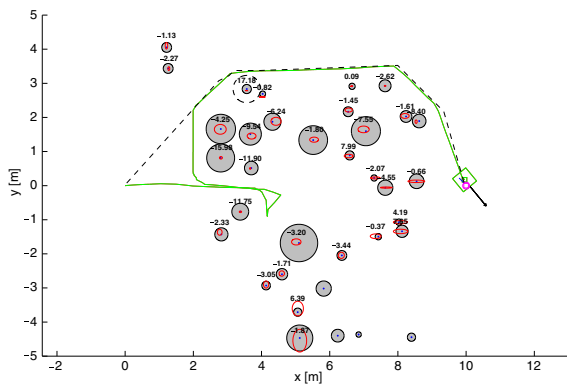


(c) An example of premature feature selection by the prime rover solo. The selected feature is in the dashed circle while three possibly better choices are shown in the dash-dotted circles.

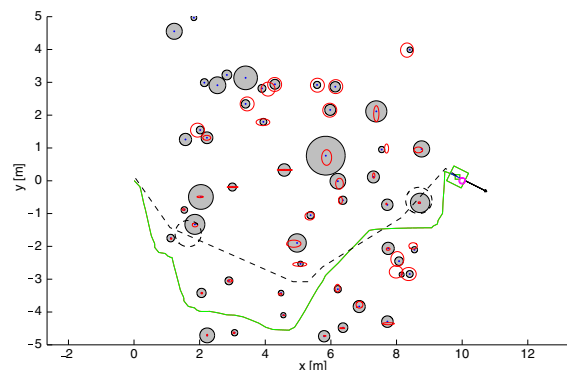
Fig. 8 Behavioural examples of the prime rover solo scenario



(a) Example of an occupancy grid generated by the scout rover. The start and goal vertices are labelled by dashed circles and the selected science vertex is labelled with an arrow. The empty vertices considered during feature selection are shaded gray.



(b) The path executed by the scout rover (solid line) prevented backtracking in the suggested path for the prime rover (dashed line).



(c) The path executed by the prime rover (solid line) significantly deviating from the recommended path provided by the scout rover (dashed line) due to the previously unobserved obstructions in the original path (shown in the dashed circles).

Fig. 9 Behavioural examples of the prime/scout rover pairing scenario

changed over the course of sequential updates, causing the rover to change its goal too frequently. Closures in the occupancy grid occurred when a feature with a high score-to-detour ratio was selected, only to later discover the path that was used to calculate the detour length was blocked. The effects of these issues could be dampened if excessive turning was penalized during path-planning.

Another issue encountered in the prime rover solo scenario was *backtracking* (Fig. 8b). In some trials, after travelling to the highest ranking feature, the rover discovered previously unobserved obstacles that blocked its path. The path planner calculated that the shortest path to the final goal required the rover to backtrack. The importance of backtracking (or other unforeseen detours) is that it can cause the violation of the constraint specified by the detour coefficient. If the detour coefficient was selected based on the power available to the rover at the beginning of the traverse, the rover may not be able to reach the final goal if the detour violation is too large. Finally, *premature feature selection* was an occasional occurrence (Fig. 8c) in the prime rover solo scenario. As the rover had no information about the features closer to the goal, the rover was often forced to visit a feature early in its traverse only to find higher scoring features later. If a large portion of the detour allowance is used to visit the first feature, higher scoring future features may need to be bypassed to ensure the final goal can be reached.

A number of the issues experienced by the prime rover solo scenario were avoided by the prime/scout rover pairing due to the availability of a feature map generated by the scout rover. This feature map allowed the prime rover to create a full plan of its traverse—including pre-selecting a feature for up-close observation. An example of this planning process is shown in Fig. 9a. Benefits of this *a priori* information included the avoidance of backtracking (Fig. 9b) and premature feature selection. However, there were also issues in the prime/scout rover pairing scenario. In some cases, an incomplete feature map provided by the scout rover made feature selection unreliable. As feature selection depends the score-to-detour ratio, the detour length might be incorrectly calculated if unobserved features block the path (Fig. 9c).

Note that “positive” adjustments are also possible if improved estimates of feature locations open up a shorter path than what was previously possible. Either way, these adjustments were not accounted for in the original feature selection, which may negatively affect the score-to-detour ratio.

5.2.2 Quantitative Comparison

In each of the thirty trials, the total distance travelled by the prime rover, the benefit score of the feature observed up-close, the score-to-detour ratio and the percentage of features that were observed was recorded for both scenarios. A quantitative comparison of the performance of the two scenarios requires the results of each individual trial to be compared, as the random feature properties, feature distribution and importance weights changed in each trial. This meant the absolute highest possible score and the absolute shortest possible path was different in each trial. Therefore, score, distance and observed features comparisons were based on percent difference from the prime rover solo results; i.e.,

$$L_{D_{\text{compare}}} = \frac{L_{D_{\text{prime/scout}}} - L_{D_{\text{primeSolo}}}}{L_{D_{\text{primeSolo}}}} \times 100 \%, \quad (11a)$$

$$S_{b_{\text{compare}}} = \frac{S_{b_{\text{prime/scout}}} - S_{b_{\text{primeSolo}}}}{S_{b_{\text{primeSolo}}}} \times 100 \%, \quad (11b)$$

$$\left(\frac{S_b}{L_D} \right)_{\text{compare}} = \frac{\left(\frac{S_b}{L_D} \right)_{\text{prime/scout}} - \left(\frac{S_b}{L_D} \right)_{\text{primeSolo}}}{\left(\frac{S_b}{L_D} \right)_{\text{primeSolo}}} \times 100 \%, \quad (11c)$$

$$(\%FO)_{\text{compare}} = \frac{(\%FO)_{\text{prime/scout}} - (\%FO)_{\text{primeSolo}}}{(\%FO)_{\text{primeSolo}}} \times 100 \%, \quad (11d)$$

where the detour length (L_D), benefit scores (S_b) and percentage of randomly generated features that were observed (%FO) comparisons are used to quantify the results. A summary of the results is presented in Table 4.

The mean values suggest that the prime rover performed better when information from a scout rover was available. The most important result, $(S_b/L_D)_{\text{compare}}$ (as this was the parameter the autonomous science algorithm was attempting to maximize), has the greatest mean and median improvement for the prime/scout rover pairing compared to the other results. The source of this advantage is apparent—the improvement in benefit score ($S_{b_{\text{compare}}}$) was much greater than the improvement in detour distance travelled ($L_{D_{\text{compare}}}$). This result indicates that the trials where the prime rover solo scenario was impaired by premature feature selection (e.g., Fig. 8c) were more common or had a greater impact than distance-based handicaps such as kinks or backtracking. The median (and maximum) results indicate the mean performance of the prime/scout rover pairing scenario was greatly influenced by trials where improvements over the prime rover solo were substantial. Once again, the improvement in these extreme trials was due to an exorbitant increase in benefit score, not in a reduction of detour distance. However, benefit score improvements must be taken with caution, as benefit scores are generated by the autonomous science algorithm itself. However, assuming sound identification, the improvement in benefit score by using a scout rover can be considerable.

One of most noticeable results in Table 4 are the wide ranges of the result parameters as indicated by the standard deviation and the minimum and maximum values. The cause of this result was the stark contrast between a nominal trial and trials where one scenario greatly outperformed

Table 4 A quantitative comparison of the two test scenarios over the course of 30 trials

Result	Mean (%)	Median (%)	St. dev. (%)	Minimum (%)	Maximum (%)
$L_{D_{\text{compare}}}$	−2.30	−0.48	13.15	−41.27	44.76
$S_{b_{\text{compare}}}$	78.38	22.23	162.87	−88.62	662.82
$\left(\frac{S_b}{L_D} \right)_{\text{compare}}$	84.47	22.66	164.07	−88.57	663.01
$(\%FO)_{\text{compare}}$	7.20	8.33	19.34	−34.09	43.75

another. Nominal trials occurred most often when a direct path to the final goal had minimal obstructions and a high-scoring feature was discovered close enough to the start position that premature feature selection by the prime rover solo was avoided. The distribution of extreme trials (e.g., trials that involved backtracking, premature feature selection, incomplete feature maps, etc.) and the nominal case likely caused the wide range of results.

6 Conclusion

This paper presents a probabilistic approach to planetary autonomous science for mobile robots. A Bayesian network is employed to estimate the probabilities that an observed feature matches a list of known objects. The attributes of features are measured with a stereo camera; however expansion to include additional sensors is possible and encouraged. The object probabilities produced by the identification process are combined with path-planning to evaluate features based on their score-to-detour ratio. Additionally, the effectiveness of pairing the prime science-capable rover with a small, rugged scout rover to aid in autonomous science is also investigated.

Laboratory experiments investigated the effectiveness of using a Bayesian network for object identification. It was found that consistency in attribute measurements and selecting attributes that allow for unique descriptions of objects were the two greatest factors in ensuring proper identification. Objects that were consistent and uniquely described tended to be well-identified, and as a result, properly scored. Poor identification of some object types could be improved by diversifying the measured attributes, improving the consistency of the machine vision algorithms, adding sensors that can measure additional object properties, and increasing the size of data sets during the training process. The evaluation algorithm was helpful in reducing the score of ambiguous identifications and distant observations. Due to the nature of observing objects using non-deterministic methods in uncertain environments, feature identification remains an unsolved

problem. However, the probabilistic approach described in this paper offers the distinct advantage of quantifying the degree of belief, and using this measure as part of the feature evaluation.

Simulated trials demonstrated how the laboratory results could be implemented into a full autonomous science system. The results indicate that the inclusion of a scout rover would be beneficial to improve the science output of rover missions using autonomous science. The transmission of a feature map from the scout to the prime rover allowed the prime rover to perform low-risk, high-benefit traverses that accounted for both science value and detour length. It was found that the *a priori* information helped the prime rover avoid pitfalls such as kinks in its path, backtracking, and exhausting its travel budget searching for science targets. In the most extreme case, the prime/scout rover pairing provided a score-to-detour ratio that was over six times greater than the prime rover solo.

6.1 Contributions

The work presented in this paper attempts to bridge together three concepts into a mutually beneficial rover exploration system, namely feature identification, feature evaluation, and path re-planning. One of the primary contributions of this paper is the observations that highlight the strengths and weaknesses of this approach. Specifically, the principal contributions can be summarized as follows:

- The development, implementation and experimental validation of a probabilistic object identification technique using BNs. The performance of this system indicates that identification performance is closely related to the quality of the instruments and algorithms providing the input data.
- A novel feature evaluation approach that combines probabilistic identification with observation quality, identification certainty, and reachability of features. The strengths and weaknesses of this algorithm is discussed at length.

- High-level observations of the benefits and pitfalls of combining science goals with general path-planning, demonstrated using simulated prime/scout rovers. Specific behaviours of rovers using this system were both qualitatively and quantitatively described. Additionally, combining autonomous science with a prime/scout rover pairing scenario is a relatively unexplored area.

6.2 Future Work

Future work could be performed with the goal of fully testing the autonomous science system on natural terrain. Some challenges would need to be addressed in achieving this goal.

- The occupancy grids used for path-planning could be augmented to cost maps, where the connections between vertices are assigned a traversal cost. This cost could be indicative of the true terrain (e.g., higher costs may be assigned to rocks, slopes and crevasses). The effect of using this map representation for detour calculations would have to be considered.
- To effectively use dynamic cost maps, a fast-replanning version of an any-angle path-planner (such as Field D* [8] or Incremental Phi* [16]) could be implemented.
- The assumption of rover localization must be replaced with an autonomous algorithm such as one of the flavours of SLAM. Additionally, data association could not be assumed. A SLAM system that includes a data association component (which could theoretically be enhanced using the object identification algorithm employed in this paper) is required to be installed. The effect of the increased uncertainty on feature selection must be analysed.
- If the prime/scout rover pairing is used, its challenges must be properly addressed. These include ensuring sufficient data transfer rates between the rovers and addressing synchronisation and power management issues.
- If the final goal of the autonomous science system is to be useful for a real planetary mission, natural features should be used, particularly those whose physical characteristics are

not unique to Earth. This effort is underway within the research group of the authors, and preliminary results in measuring the attributes of natural targets are encouraging.

Appendix A: Simulation Parameters

The simulation parameters are outlined in Table 5. These parameters were kept constant for each trial of the simulated tests. One exception is the importance weights, which were each assigned a random value in the uniform distribution $U(-1, 1)$.

Appendix B: Laboratory Experiment Objects

Figure 10 shows the 13 unique objects that were used in the laboratory experiments. Note that the white plastic ball and the yellow plastic ball (Fig. 10h and i, respectively) differ only in colour, as do the small green pot and small brown pot (Fig. 10j and k, respectively). The standard die in the images has an edge length of 16 mm.

Table 5 The simulation parameters

Parameter	Value
Camera field-of-view	66°
Camera range	5 m
Desired forward velocity	2.2 cm/s (80 m/h)
Detour coefficient (D)	1.5
Distance between waypoints	0.25 m
Dynamic benefit score threshold multiplier (m_D)	$(\mu + \sigma)/\mu$
Importance weights (a, b, c, d)	$U(-1, 1)$
Maximum dimension of rover footprint (h)	0.5 m
Maximum rover angular velocity	3.63 °/s
Maximum rover forward velocity	2.2 cm/s (80 m/h)
Minimum rover angular velocity	-3.63 °/s
Minimum rover forward velocity	0 cm/s
Occupancy grid resolution	0.15 m
Scale constant (K)	100
Science sensor radius (r_{sen})	0.50 m
Static benefit score threshold ($S_{b,\text{stat}}$)	0
Time step (T)	0.25 s

Fig. 10 The objects used for the laboratory experiments



(a) Pink volleyball (PVB).



(b) White volleyball (WVB).



(c) Small soccer ball (SSB).



(d) Large soccer ball (LSB).



(e) Bowl (BWL).



(f) Small football (SFB).



(g) Large football (LFB).



(h) White plastic ball (WPB).



(i) Yellow plastic ball (YPB).



(j) Small green pot (SGP).



(k) Small brown pot (SBP).

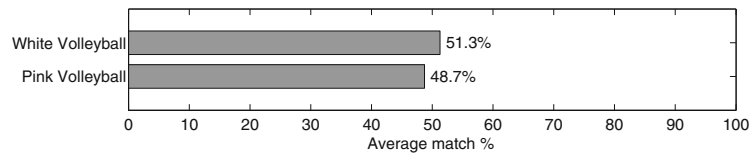


(l) Medium pot (MPT).

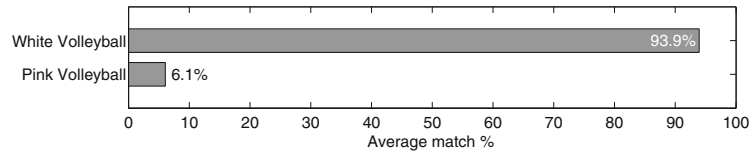


(m) Large pot (LPT).

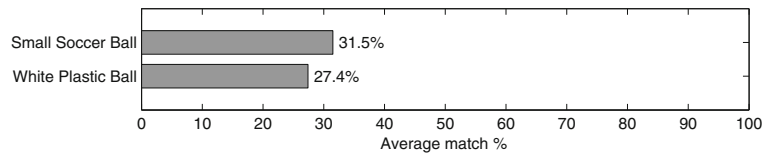
Fig. 11 The top two average match percentages by the Bayesian network for each of the 13 objects used in the laboratory experiments



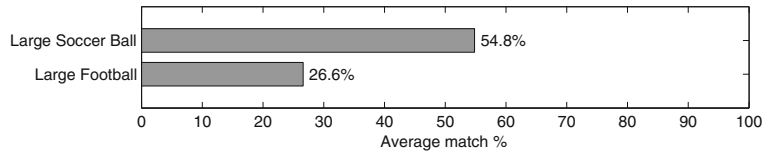
(a) Pink volleyball (4 detections).



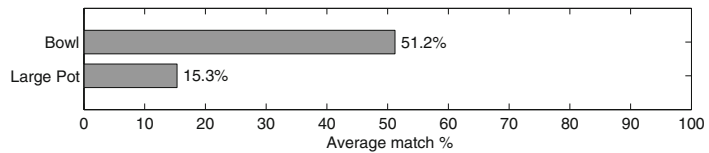
(b) White volleyball (3 detections).



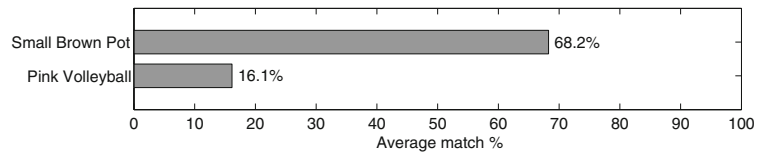
(c) Small soccer ball (3 detections).



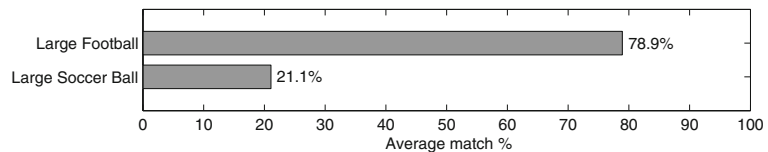
(d) Large soccer ball (11 detections).



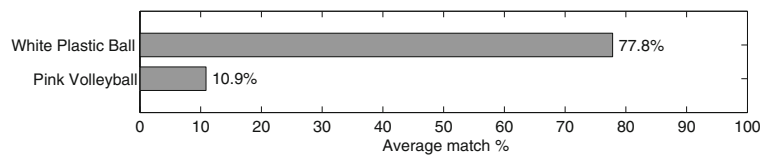
(e) Bowl (22 detections).



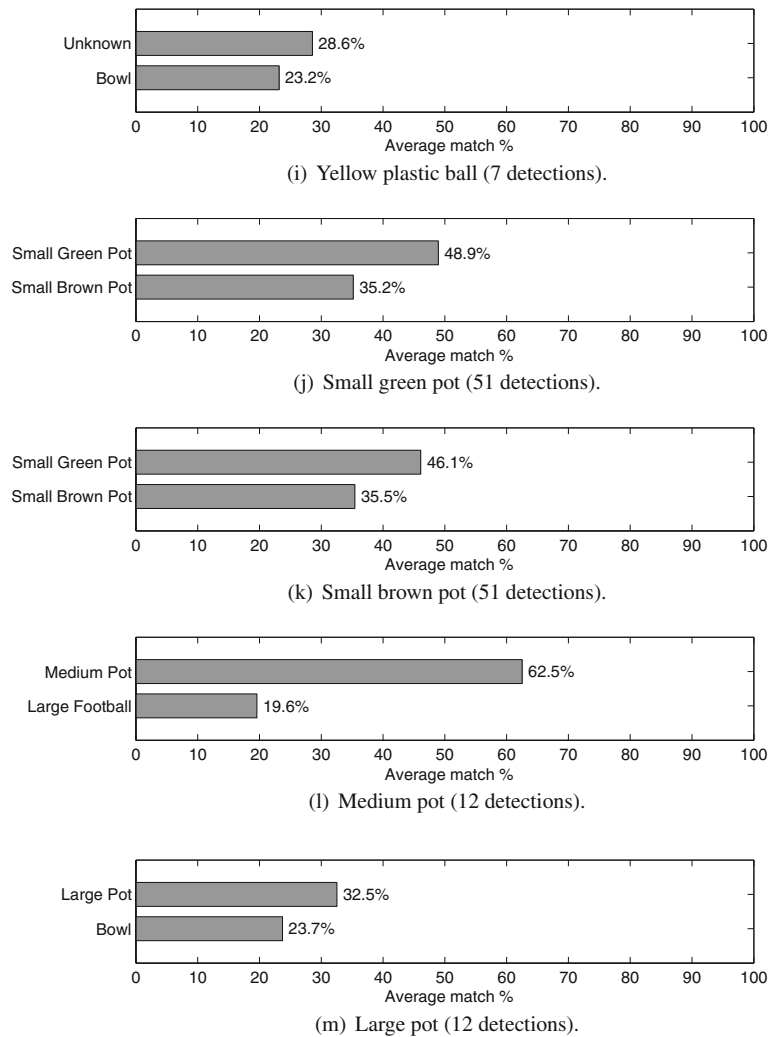
(f) Small football (4 detections).



(g) Large football (5 detections).



(h) White plastic ball (9 detections).

Fig. 11 (continued)

Appendix C: Laboratory Experiment Match Percentages

The top two match percentages assigned by the Bayesian network in the laboratory experiments for each of the 13 objects are illustrated in Fig. 11.

Appendix D: Laboratory Experiment Feature Map

The feature map that was generated by the rover during the first trial is shown in Fig. 12. The poses of the rover from which images were captured are the arrowed boxes labelled in alphabetical order.

Circles represent the feature locations, with the name of the object that occupied each feature location printed above. Objects that were detected by the machine vision algorithm are marked with an “X” and have their benefit bias scores printed below. As automatic data association has not yet been implemented (see Section 2.1), it was performed manually. When a feature was observed multiple times, the score it was assigned when its apparent size was greatest was used. This represented the occasion when the most visual information about the feature was available. The benefit bias equation that was used to calculate benefit scores used the general form reported in (2) using the importance weights outlined in Table 2.

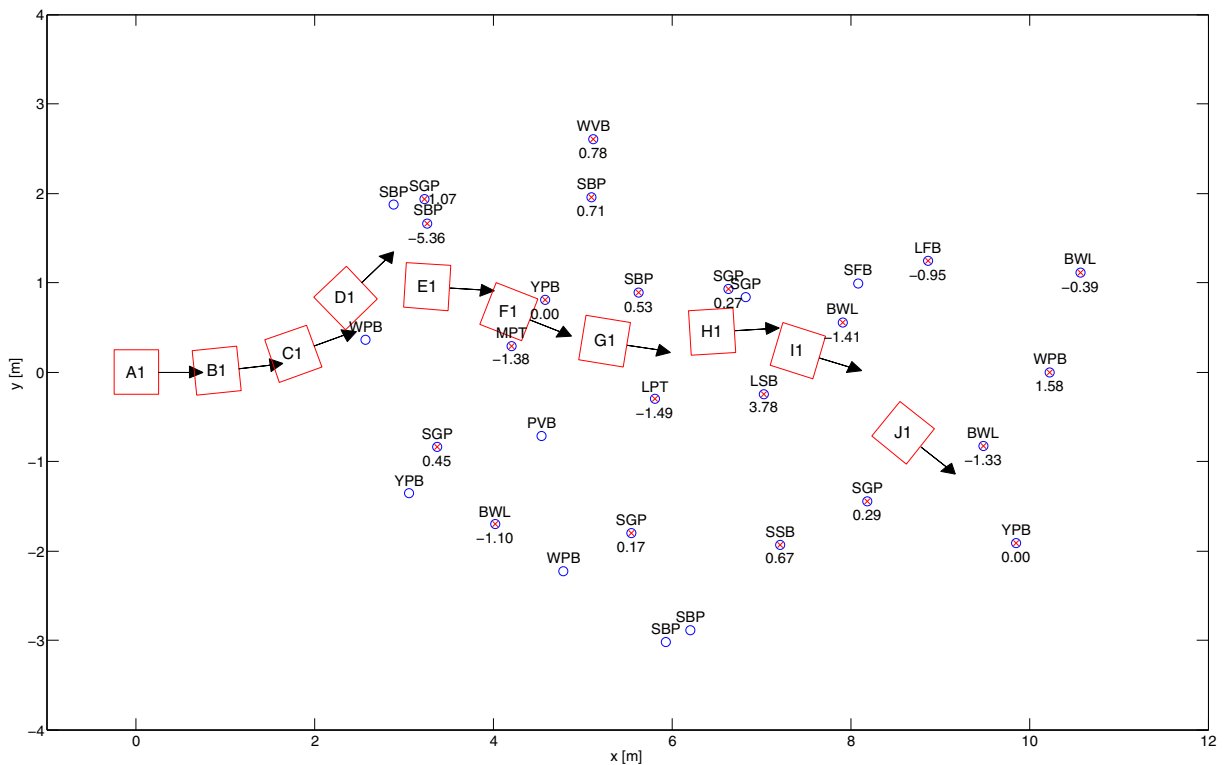


Fig. 12 Path and scoring results for the first laboratory trial

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